

The arc is fine above 4 kHz. Below 2 kHz the room bends it by up to 8 dB.

For these runs the arc was laid flat (horizontal), with the propeller axis vertical. All microphones then sit at the same height, in the propeller plane, around the propeller. A perfect setup reads the same level at every position: a circle. We did this twelve times (keys prop7–18, 1–2 Sept 2026, PWM 1900). What is left after we remove each microphone's own offset is the effect of the room.

2026-09-08
data: laptop backup of 2 Sept
script: scripts/arc_error_map.py
full guide: docs/arc-validation-guide.html

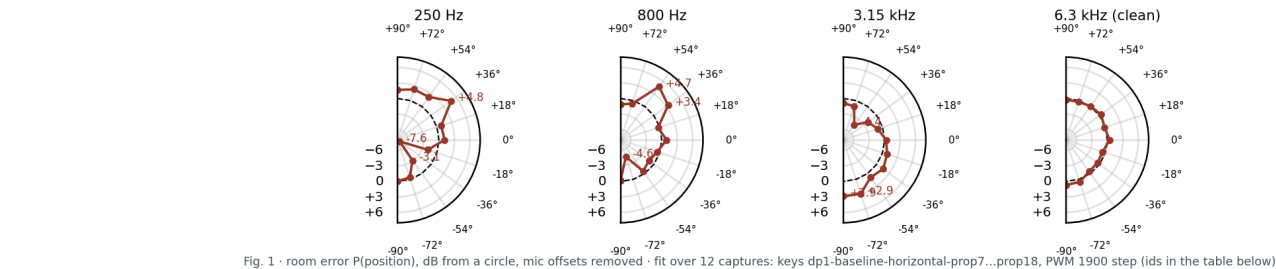


Fig. 1 · room error P(position), dB from a circle, mic offsets removed · fit over 12 captures: keys dp1-baseline-horizontal-prop7...prop18, PWM 1900 step (ids in the table below)

What the room does (right panel of Fig 2)

How to read the map. Row = spot on the arc, column = frequency band. White: the spot reads what a circle predicts. Red: too loud. Blue: too quiet. Each microphone's own error is already taken out (left panel), so the colour is the room.

Deep blue at -72° and red at $+36^\circ/+54^\circ$, 500-1000 Hz. A sound wave that bounces off a nearby surface arrives a little later than the direct wave. If the extra path is half a wavelength the two cancel and the spot goes blue; if it is a whole wavelength they add and the spot goes red. The extra path depends on the spot, so neighbours can be opposite colours and one spot can flip colour from band to band. Wavelength here is 35–70 cm, so the surface is within about a metre.

The 200-250 Hz column: the blade tone. -36° is deep blue, $+36^\circ$ red. Wavelength about 1.4 m: only a large surface (wall, floor, table) reflects a wave that long. The blue cell moves with motor speed because the tone slides through the cancelling frequency, the textbook sign of a reflection.

One side blue, the other red. Every deep blue cell is at a negative angle (-36° , -54° , -72°), every strong red at a positive angle. A surface on one side of the rig does exactly this.

White above 4 kHz. Short waves are absorbed by the foam and scattered by small objects, so only direct sound reaches the microphones. The arc itself is fine.

What it is not

Not the microphones, with one exception. Because the microphones at $+72^\circ$ and $+54^\circ$ were swapped after prop10, and the arc was turned around for prop11–14, we can tell apart what belongs to a microphone and what belongs to a position. Ten microphones agree within ± 1.8 dB in every band. **Microphone 811-1892 reads 2-6 dB too low from 125 Hz to 2.5 kHz.** Its calibration file does not match the capsule (its sensitivity value, -10.63 dB, is the least negative of the set, while the raw capsule is only about 1 dB low). When it sat at $+54^\circ$ it made the room error look like 7 dB. Take it off the arc or measure it again.

Not a labelling error. prop11, 12, 13 and 14 were measured with the arc turned around. Their patterns match the mirror image of the other runs (correlation $+0.79$ to $+0.98$). Plotted by label, the dip appears on the other side. Plotted by physical position, they agree with the other eight runs.

Not one frequency. The error sits in three separate bands, 200–250 Hz, 630–1000 Hz and 3150 Hz, and nowhere else (Fig. 2). Rajmane and Baumann (DAGA 2016) found a steel plate in a chamber only shows up once the wavelength is shorter than the plate (35 cm from 1.25 kHz, 10 cm from 3.15 kHz). So three bands point to three object sizes: about 1.5 m (floor, table or wall), about 0.4 m (stand base or a bar of the arc frame), about 0.1 m (a microphone clamp or mount).

Fig. 2 · Microphone error map (left) and room error map (right), same fit, same scale

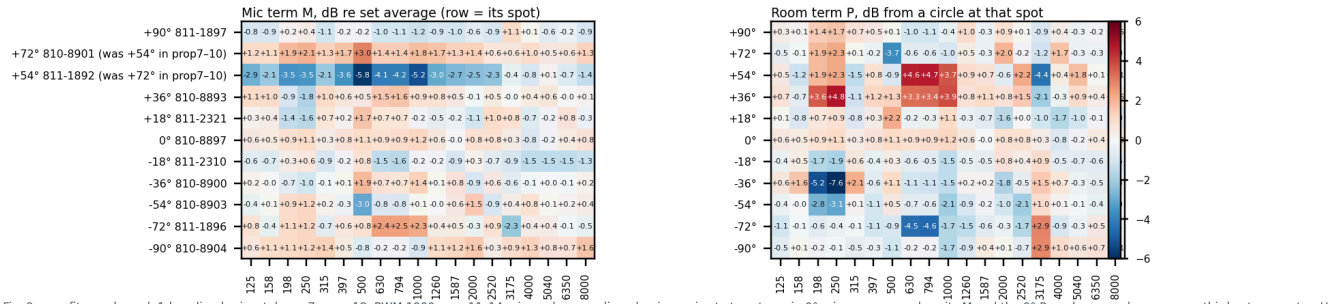


Fig. 2 · one fit over keys dp1-baseline-horizontal-prop7...prop18, PWM 1900, prop11-14 mirrored · rows aligned: mic row i sat at spot row i · 0° mic never moved, so its M and the 0° P are known only as a sum · third-octave centre, Hz

Where it points

With the arc flat, all microphones are at the same height. Floor and ceiling reflect the same into every microphone and cannot make one half of the arc differ from the other. The pattern we see is one-sided: the $-36^\circ/-54^\circ/-72^\circ$ side reads low, the $+36^\circ/+54^\circ$ side reads high. So the surface is beside the propeller, in the horizontal plane: the wall, the corner or an object on the $-$ angle side of the rig. The dip at 240 Hz that moves with motor speed means the bounced path is about 0.7 m longer than the direct one (half a wavelength), so that surface is close, within about half a metre of the microphones. The -72° dip at 630–1000 Hz (wavelength about 0.4 m) is a smaller object at the -90° end of the arc, on the corner side.

Next test (one afternoon)

- Take 811-1892 off the arc.
- Put a small loudspeaker on the stand instead of the propeller. Play short tone bursts at 250 and 800 Hz and record all microphones. The delay of the second arrival at each microphone, times 343 m/s, is the extra path length. Only one surface fits all eleven delays. With no airflow, anything left is the room.
- Write down the geometry in docs/mic_arc.md: arc radius, propeller height, which wall or corner is at the -90° end, distances from the -72° and $+72^\circ$ microphones to the nearest wall.
- Cover or move that surface, then run scripts/arc_error_map.py again. Foam only helps at 3 kHz. At 250 Hz the only fix is distance.

How the numbers were made (not a sum, not an average)

Every level in the twelve runs (11 microphones $\times$ 19 third-octave bands $\times$ 12 runs) is written as three parts added together: **run gain + position error + microphone offset**. The three parts are found by least squares, i.e. the single set of values that fits all 12×11 levels best in every band. **Run gain** takes up any level difference between runs (RPM, day, load). **Microphone offset** takes up each serial's own error after calibration. **Position error** is what is left that belongs to a spot on the arc; it is forced to average zero over the eleven spots, so it reads as dB from a circle. That is what Figs. 1 and 2 show.

The turned-around runs are handled before the fit: for prop11–14 the label +e is entered as physical $-e$, so -72° always means the same spot in the room. The four microphone offsets and the eleven position errors can be told apart only because the same spot was covered by different microphones (the swap after prop10 and the four turned runs). Without that, a low microphone and a low spot would be indistinguishable. Spectra: Welch 4096-point Hann, UMIK-2 calibration as the app applies it, third-octave sums 125 Hz–8 kHz. Data: the last PWM-1900 capture in each key 2004_6in_unset_dp1-baseline-horizontal-prop7...prop18, 11 mics $\times$ 2.0 s; prop11–14 were measured with the arc turned around; prop1–6 and the 31 Aug keys are left out because they match neither orientation. Capture ids per key: docs/analysis/arc-validation-data.json. Code: scripts/arc_error_map.py, function fit.

Do the numbers add up?

Refitting from the data pack reproduces the published maps to 0.01 dB; each map sums to zero per band by construction. 132 measured cells per band are explained by 34 unknowns, 87% of all cells are within 2 dB of gain + P + M. Scatter is 0.6–0.8 dB in the quiet bands and 1.5–2.3 dB in 200–250 Hz and 500–1000 Hz, the bands where the room acts. The worst cells ($+11$ dB at $-36^\circ/250$ Hz in prop7; $+6$ to $+7$ dB at $0^\circ/2-4$ kHz in prop7–8) are single tone-band cells in the first two runs where the blade tone sits on a band edge; they do not follow a microphone or a spot and move the maps by under 1 dB. Limit: the 0° microphone (810-8897) never moved, so its M and the 0° P are known only as a sum.

Worked example · -72° spot at 800 Hz, two mics, two orientations

RUN	MIC THERE	MEASURED	GAIN	P	M	MODEL	LEFT
prop18	811-1896	59.1	62.3	-4.6	+2.5	60.2	-1.1
prop12 (180°)	810-8901	58.6	62.2	-4.6	+1.4	59.0	-0.4

$+54^\circ$ at 800 Hz: prop8 with 810-8901 reads $+5.6$ re gain = $P + 4.7 + M + 1.4$; prop18 with 811-1892 reads $+3.1 = +4.7 - 4.2$. Two mics, one room value.